



Curitiba - Parana - Brazil
+55 41 98868-5818
marcelo.feliciano.f@outlook.com

Marcelo Feliciano Filho

Technical Team Lead | Senior Software Engineer (IC4) | Python | Django | PostgreSQL | Docker | RAG | MCP | PhD Student at PUCPR/PPGEPS | Master's Degree in Machine Learning and Systems Engineering

ABOUT

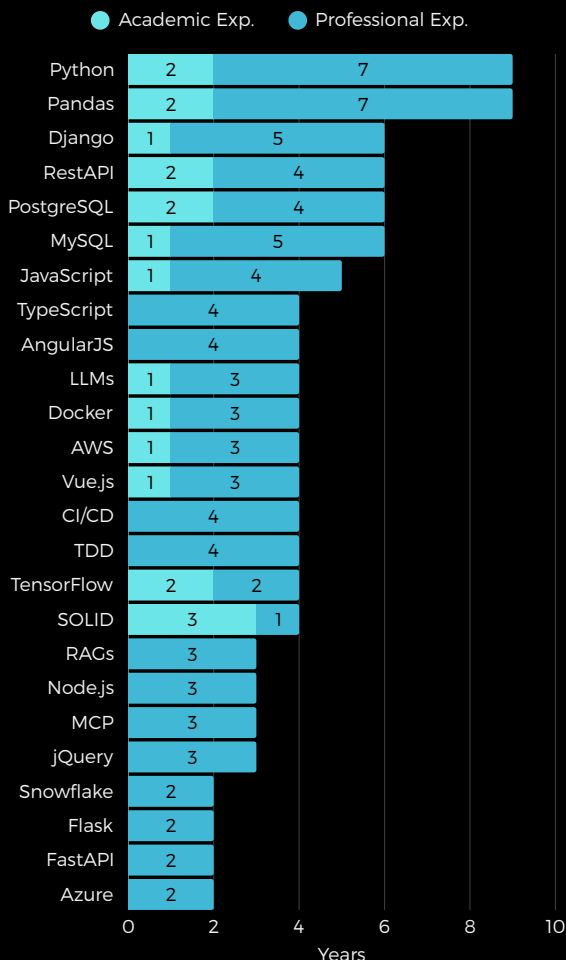
Senior Software Engineer and Tech Lead with 7+ years building scalable backend systems and AI-driven pipelines in Python. Currently pursuing a PhD in Artificial Intelligence at PUCPR, with published research in smart manufacturing and human-centric systems. Specializes in RAG architectures, microservices, and bridging applied research into production-grade infrastructure.

PROFESSIONAL EXPERIENCE

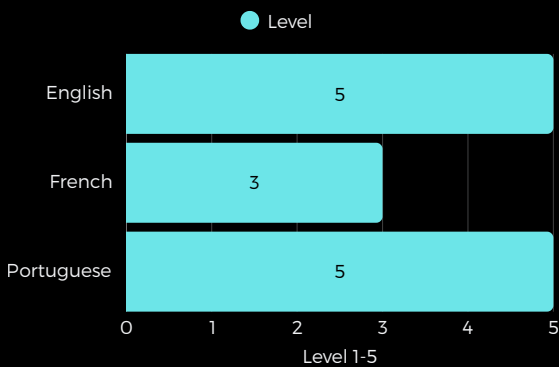
- **BT (British Telecommunications) - Remote (London, UK)**
 - **Senior Software Engineer - Sep/2025 ~ Current Job**
 - Resolve complex production issues (idempotency failures, scalability bottlenecks, device misconfigurations) across BT's core NaaS platform.
 - Deliver features, bug fixes, and pipeline improvements using Django, Pydantic, Docker, and CI/CD.
 - Write BDD, unit, and feature tests following TDD practices; implement SOAK tests against real field devices.
- **TURING - Remote (Los Angeles County, CA)**
 - **TECHNICAL TEAM LEAD - Dec/23 ~ Sep/2025**
 - Architected a production RAG pipeline (FastAPI, LangChain, pgvector) for clinical document querying, reducing hallucination rate ~70% via hybrid retrieval and cross-encoder reranking
 - Integrated MCP to expose retrieval and feature-lookup operations as typed LLM tools, decoupling orchestration from business logic.
 - Deployed microservices on AWS (EC2, ECS, S3) with Docker/CI-CD.
 - Improved product feedback rating 40% through RAGAS-based evaluation (faithfulness + relevance) against a labeled QA dataset.
 - Led and mentored 6-8 engineers on LLM integration, async Python, and RAG evaluation practices.
 - **SENIOR SOFTWARE ENGINEER - Aug/2022 ~ Dec/2023**
 - Delivered backend engineering solutions for multiple clients using **Python**, Django/FastAPI, Snowflake, PostgreSQL, and Docker. Managed deployments via AWS (LightSail, EC2, S3) and supported frontend integration using Angular.js and Vue.js. Contributed to full-stack application performance and architectural decisions in cloud-native environments.

HARD SKILLS

Over 7 years of combined academic and professional experience specializing in backend development with Python and Django, with deep expertise in building scalable APIs and data-driven systems using PostgreSQL and MySQL. Proficient in designing and implementing RAG (Retrieval-Augmented Generation) pipelines and integrating Model Context Protocol (MCP) for intelligent, context-aware backend services. Experienced with modern backend frameworks such as FastAPI and Node.js, and well-versed in DevOps practices including Docker, CI/CD automation, and deployment on AWS and Azure. Strong foundation in software engineering principles (TDD, SOLID) and optimizing backend performance for AI-enabled and cloud-native applications.



LANGUAGES



SOFT SKILLS

Strong analytical thinker with excellent logical reasoning and a passion for continuous learning. Known for approaching challenges with creativity, adaptability, and a solution-oriented mindset to drive innovation and team success.

EDUCATION

- Doctor of Philosophy (PhD), Artificial Intelligence - 2024 ~ Current
 - Pursuing a PhD focused on hybrid AI for Industry 5.0.
- Master of Engineering (MEng), Systems Engineering - 2021 ~ 2023 | GPA: A
 - Completed a master's degree emphasizing Artificial Intelligence. Gained hands-on experience with FastAPI, REST APIs, and cloud-based model development using tools like Python, PyTorch, and Simulink.
- Bachelor of Engineering (BE), Control and Automation Engineering, 2016 ~ 2020 | GPA: 8.4
 - Graduated with honors, receiving the Marcelino Champagnat Award for top academic performance. Focused on control systems, automation, and intelligent software design using Python, MATLAB, Simulink, and machine learning tools.

HONORS & AWARDS

- **Marcelino Champagnat – Issued by (PUCPR) – Feb 2021**
 - Awarded for graduating with the highest academic performance in the Control and Automation Engineering program.

CONTACT ME



- **DAYWAY - SOFTWARE ENGINEERING II - Apr/2022 ~ Aug/22**
 - Developed clean, scalable backend systems and cloud integrations using modern software practices. Collaborated with cross-functional teams to enhance infrastructure and ensure robust deployment pipelines through CI/CD and task queues. Applied machine learning and data-driven insights to improve solution accuracy and maintainability in Python.
- **LACTEC - SOFTWARE ENGINEER - Sep/21 ~ Apr/22**
 - Built an EV recharging platform from scratch (Django, Angular, PostgreSQL, Snowflake, Docker) with Stripe payment processing. Led sprint planning, code reviews, and team collaboration using SOLID principles and agile.
- **ATELIWARE - SOFTWARE ENGINEER - Jul/21 ~ Sep/21**

Delivered features across multiple web projects with financial transactions (**STONE STORE**) using Ruby on Rails, Elixir Phoenix, Docker, and AWS.
- **AI ENGENHARIA**
 - **Software Developer - Dec/2020 ~ Jul/2021**
 - Optimized AI-based document handling systems, cutting registration time by 95% and document comparison time by 80%. Stack: Django, MySQL, JavaScript.
 - **Junior Web Developer - Aug/2019 ~ Dec/2020**
 - Developed engineering solutions in a collaborative environment using GitLab and TDD practices.

ACADEMIC EXPERIENCES / INTERNSHIPS

- **TECH INITIALIZATION PROJECT - PIBITI - Jul/2018 ~ Jul/2019**

Built an electronic model of Curitiba's Innovation Ecosystem using Unity, Python, and C#. Automated data collection and simulation behaviors, translating academic frameworks into interactive digital representations.
- **SCIENTIFIC INITIALIZATION - PIBIC - Jun 2016 ~ Jun 2018**

Co-developed a smart factory prototype aligned with Industry 4.0 at PUCPR's Systems Lab. Integrated RFID readers, PLCs, and a UR5 collaborative robot via Modbus protocols. Co-authored a paper published at FAIM (Modena, Italy).

PUBLICATIONS

- 1.M. F. Filho, P. A. Boareto et al. — International Journal of Production Research, 2026. DOI: [10.1080/00207543.2026.2719846](https://doi.org/10.1080/00207543.2026.2719846)
- 2.M. F. Filho, A. L. Szejka, E. R. Loures. "Human-Centric Smart Manufacturing: A Framework Utilizing Digital Twins and Cyber-Physical Production Systems." Int. Conference on Innovative Intelligent Industrial Production and Logistics, 2024. (3 citations)
- 3.M. Feliciano Filho, Y. Liao, E. R. Loures, O. C. Junior. "Self-aware Smart Products: Systematic Literature Review, Conceptual Design and Prototype Implementation." Procedia Manufacturing 11, 1471-1480, 2017. (10 citations)
- 4.G. Reynoso-Meza, V. H. A. Ribeiro, M. Feliciano Filho, B. Z. de Abreu. "Time Stacking Decision Tree for Oscillatory Failure Case Detection in a Flight Control System." IFAC-PapersOnLine 56(2), 354-359.